

Amjad S Al-thabteh

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Education

B.S in Computer Science GPA: 3.65/4.00

Wayne State University

Detroit

May

Experience

SWE Intern

Immersible Graphics

Detroit, MI

Sep 2025 – Dec 2025

- Architected a real-time C++20/OpenGL rendering engine with GPU-driven pipelines and modular subsystem boundaries.
- Built JSON-driven asset-authoring tools and internal UI, improving content iteration flexibility across teams.
- Implemented data-driven rendering optimizations enabling runtime analysis and AI-guided heuristics to balance visual quality.
- Modular rendering subsystems with clear ownership boundaries to support scalability and long-term feature growth
- Led the design and delivery of a GPU-accelerated rendering engine to 40+ cross-functional users, incorporating feedback.

Software Development Intern

Rare Munchies

Ann Arbor, MI

Sep 2023 - Apr 2024

- Integrated UI components using TypeScript, improving page transitions and interaction flow to enhance user experience.
- Resolved **cross-browser compatibility issues**, reducing debugging time and unblocking delivery
- Improved UI responsiveness and consistency across modern browsers, stabilizing production releases.

Projects

Glint 3D –3D Rendering Engine

[Glint Render](#)

- Engineered a real-time C++17/OpenGL rendering pipeline with dynamic GPU workload balancing and lighting optimization.
- Designed a cross-platform modular architecture separating rendering, scene management, and resource systems.
- Pushed Runtime scene analysis to maintain stable frame performance under complex object and lighting conditions
- Implemented a custom memory pool and allocator system using C++20, significantly reducing heap allocations.

UE -5.6 Cinematic “No One Answers Clip”

[No one answers](#)

- Developed a solo cinematic vertical slice in Unreal Engine 5.6 using Metahuman animation, cinematic cameras, Control Rig.
- Used Niagara effects and environmental animation to tell the story visually and set the tone for No One Answers
- Structured as a production-ready prototype for a narrative-driven detective game.

High-Frequency Trading Engine (C++20)

- Built a low-latency C++20 limit order book with price-time FIFO matching, spread detection, and trade aggregation.
- Designed cache-efficient data structures and custom timing instrumentation to benchmark microsecond-level execution latency.

C++ Runtime Call Stack & Diagnostics Engine (TraceForge)

- Developed a C++23 **debugging system** that performs automated stack-trace parsing.
- Implemented automated regression test generation and rule-based fix suggestions derived from stack trace patterns
- Added **thread-safe logging** to capture errors and runtime events in multithreaded applications.
- **Memory tracking utilities** to help identify leaks and invalid allocations during debugging.

UE 5.5 RPG

- Wrote gameplay systems **using (C++ 20 and Visual Scripting)**, connecting animation states, environmental triggers.
- Created 3D environments and characters from scratch, rigging for animation.
- Produced a 90+ second real-time cinematic using MetaHumans, Control Rig, and Sequencer, designing scene flow and timing.

Skill

- **Programming Languages:** C++ (11–23), C#, C, Python (3.7 - 3.12), TypeScript, Java
- **Graphics & Engines:** OpenGL, Unreal Engine 4–5.7, Unity, NX Design
- **Course Work:** Computer Graphics, Game Programming, **Software Engineering**, **Operating Systems**, Linear Algebra
- **Specialties:** Gameplay Systems, VR/AR Development, Simulation, Tools & Plugins, Technical Art